

Fundamentals of Python



Objectives

- Discuss the importance of Python
- Provide the steps and ways of installing and using Python.
- Demonstrate how to code in Python.

What is Python

- Python is a popular programming language. It was created by Guido van Rossum, and released in 1991.

It is used for:

- web development (server-side),
- software development,
- mathematics,
- system scripting.

Capabilities of Python

- Python can be used on a server to create web applications.
- Python can be used alongside software to create workflows.
- Python can connect to database systems. It can also read and modify files.
- Python can be used to handle big data and perform complex mathematics.
- Python can be used for rapid prototyping, or for production-ready software development.

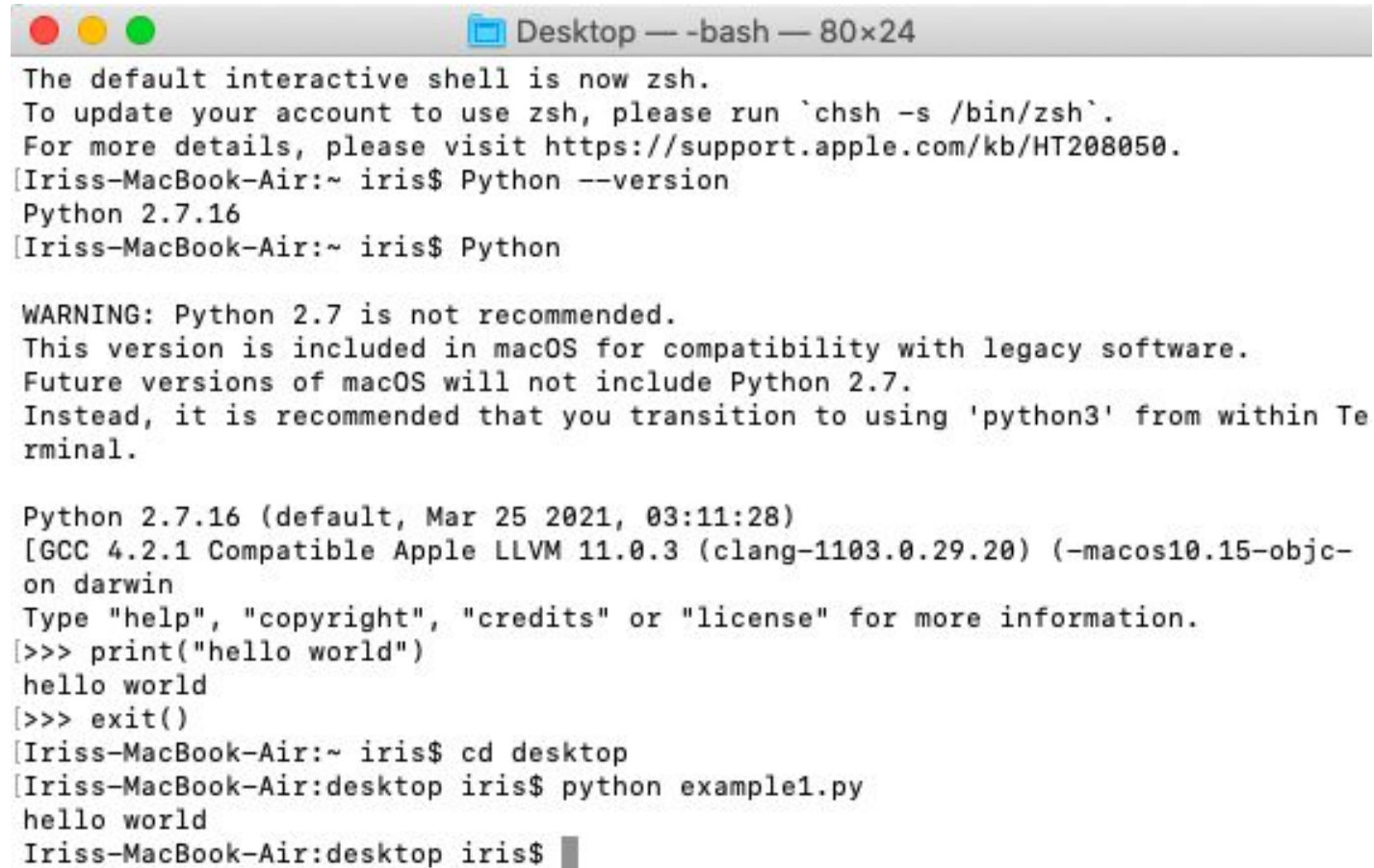
Python Syntax compared to other programming languages

- Python was designed for readability, and has some similarities to the English language with influence from mathematics.
- Python uses new lines to complete a command, as opposed to other programming languages which often use semicolons or parentheses.
- Python relies on ***indentation, using whitespace***, to define scope; such as the scope of loops, functions and classes.
- Other programming languages often use curly-brackets for this purpose.

Using Python

- Know the version of Python using the command line (Windows).
 - Just type: `Python --version`
- Access it using your command line.
 - Just type: `Python` (you will be inside the Python shell, and you can start coding)
- You can use notepad or any text editor, type the code, and run it using the command line. You must save the text file as `.py`

Using Python



A terminal window titled "Desktop — -bash — 80x24" with standard macOS window controls (red, yellow, green buttons). The terminal output shows the following sequence of commands and responses:

```
The default interactive shell is now zsh.  
To update your account to use zsh, please run `chsh -s /bin/zsh`.  
For more details, please visit https://support.apple.com/kb/HT208050.  
[Iriss-MacBook-Air:~ iris$ Python --version  
Python 2.7.16  
[Iriss-MacBook-Air:~ iris$ Python  
  
WARNING: Python 2.7 is not recommended.  
This version is included in macOS for compatibility with legacy software.  
Future versions of macOS will not include Python 2.7.  
Instead, it is recommended that you transition to using 'python3' from within Te  
rminal.  
  
Python 2.7.16 (default, Mar 25 2021, 03:11:28)  
[GCC 4.2.1 Compatible Apple LLVM 11.0.3 (clang-1103.0.29.20) (-macos10.15-objc-  
on darwin  
Type "help", "copyright", "credits" or "license" for more information.  
[>>> print("hello world")  
hello world  
[>>> exit()  
[Iriss-MacBook-Air:~ iris$ cd desktop  
[Iriss-MacBook-Air:desktop iris$ python example1.py  
hello world  
[Iriss-MacBook-Air:desktop iris$
```

Installing Python



Python.org

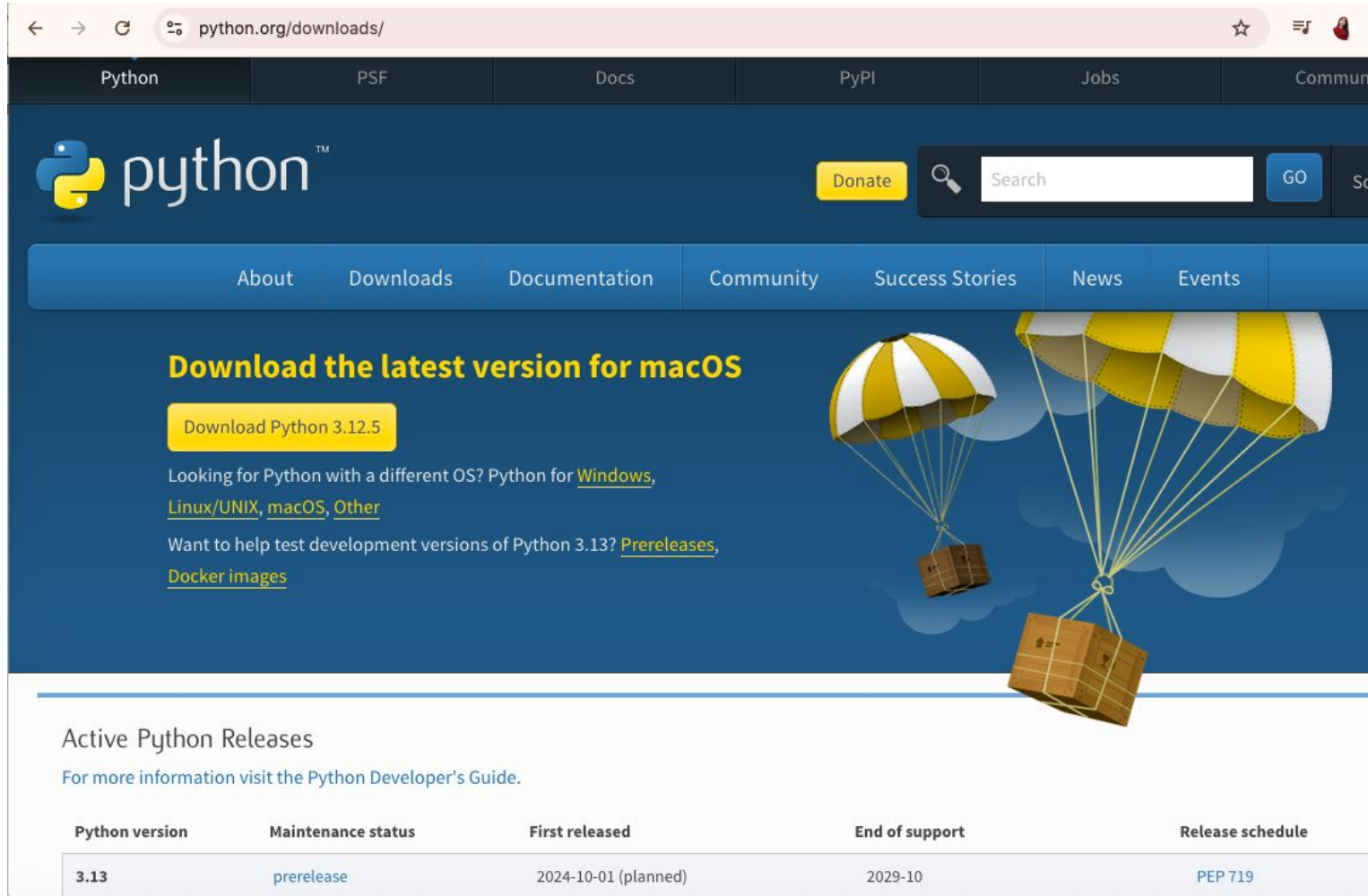
<https://www.python.org> › [downloads](#) ⋮

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[Python Releases for Windows](#) · [Python 3.12.4](#) · [Python Releases for macOS](#)

Installing Python

A screenshot of the Python.org/downloads page. The browser address bar shows 'python.org/downloads/'. The page has a dark blue header with the Python logo and a navigation menu. Below the header is a blue banner with the text 'Download the latest version for macOS' and a yellow button 'Download Python 3.12.5'. To the right of the button is an illustration of two parachutes carrying boxes. Below the banner is a section titled 'Active Python Releases' with a link to the Python Developer's Guide. At the bottom is a table with columns: Python version, Maintenance status, First released, End of support, and Release schedule. The table has one row for Python 3.13.

python.org/downloads/

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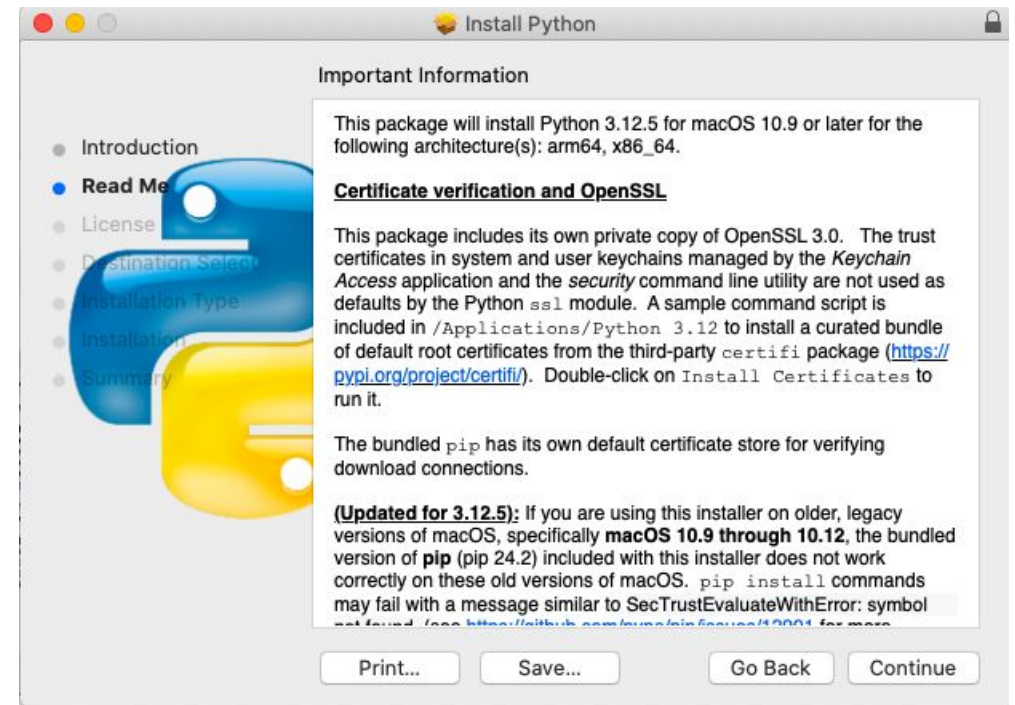
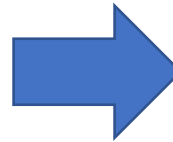
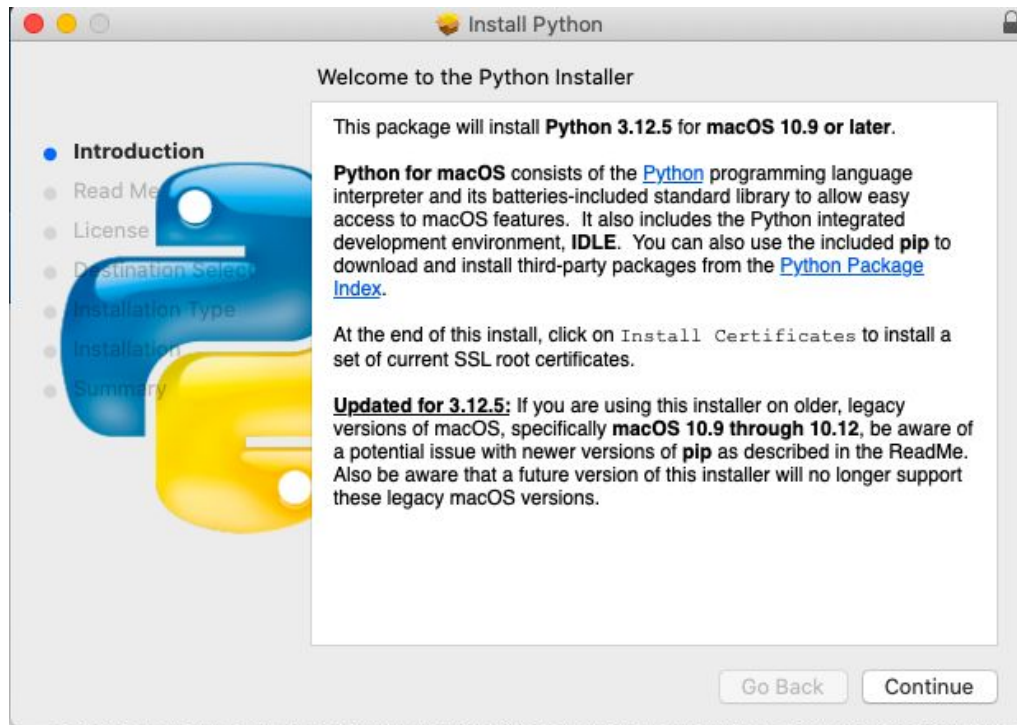
Want to help test development versions of Python 3.13? [Prereleases](#), [Docker images](#)

Active Python Releases

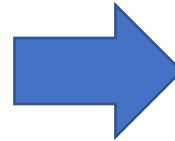
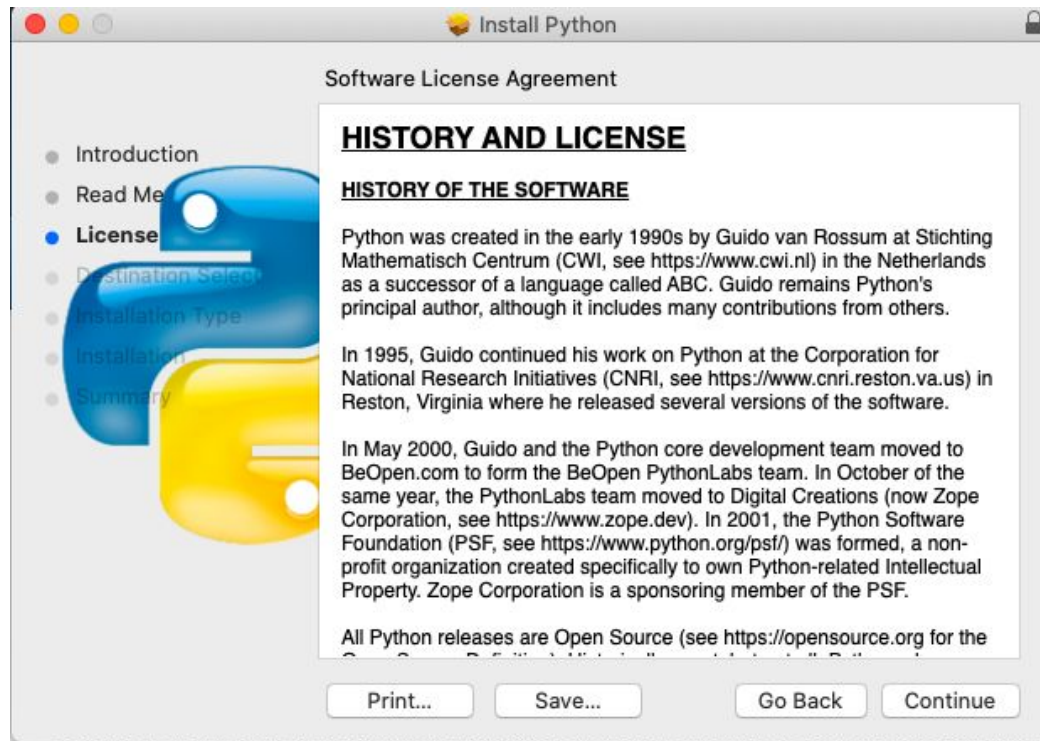
For more information visit the [Python Developer's Guide](#).

Python version	Maintenance status	First released	End of support	Release schedule
3.13	prerelease	2024-10-01 (planned)	2029-10	PEP 719

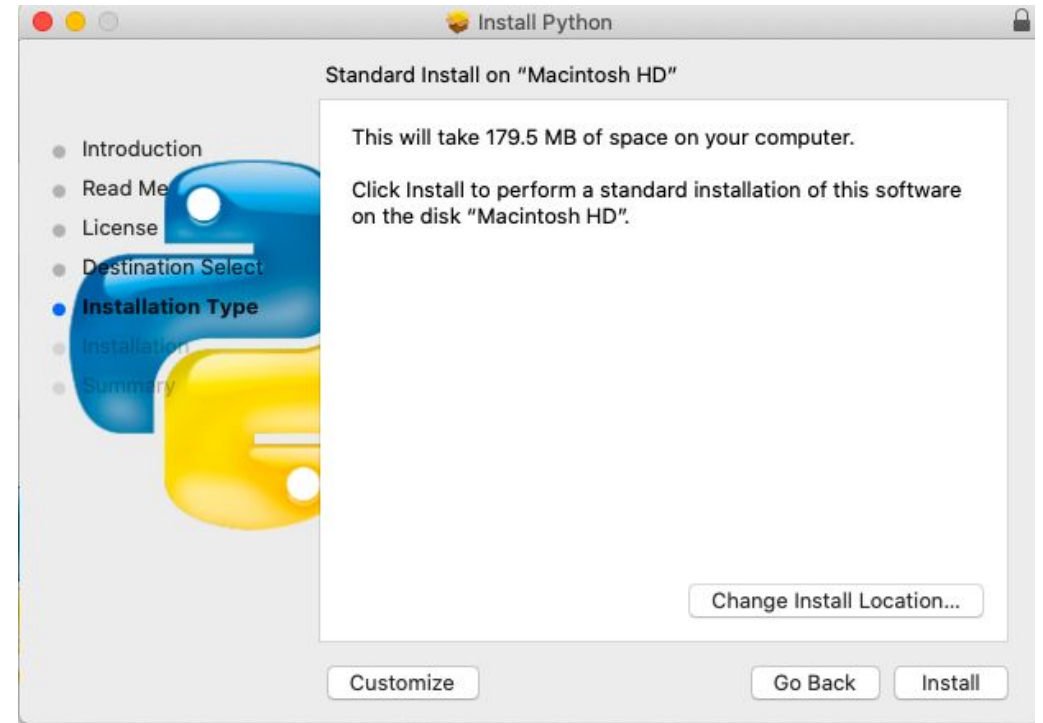
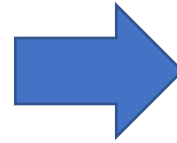
Installing Python (Mac)



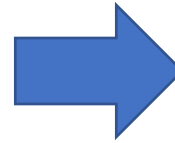
Installing Python (Mac)



Installing Python (Mac)



Installing Python (Mac)



Fundamentals of Python- Examples

▶ #Example 1

```
print ("Hello World")
```

⇒ Hello World

[] #Example 2

```
#This is a single line comment  
'''This is a multiple  
line  
comment '''
```

⇒ 'This is a multiple\nline\ncomment '

[] #Example 3

```
print ("Hello World " *5)
```

⇒ Hello World Hello World Hello World Hello World Hello World

Fundamentals of Python- Examples

```
[ ] #Example 4
a=5
b=7
c=a+b
print ("c = ", c)

d='x'
e='y'
print (d+e)

f="Hello"
g=" World"

l= f + "" + g
print (l)
```

⇒ c = 12
xy
Hello World

```
[ ] #Example 5
a = 200
b = 100
if b > a:
    print("b is greater than a")
```

```
[ ] #Example 6
for i in range (6):
    print (i)
    print("Finished")
```

⇒ 0
Finished
1
Finished
2
Finished
3
Finished
4
Finished
5
Finished

Fundamentals of Python- Examples

```
[ ] #Example 7
    for i in range (5):
        for j in range (5):
            x=i*j
            print(x)
        print("Inner Loop Finished")
    print ("Outer Loop Finished")
```

```
⇒ 0
   0
   0
   0
   0
   0
   Inner Loop Finished
   0
   1
   2
   3
   4
   Inner Loop Finished
   0
   2
   4
   6
   8
   Inner Loop Finished
   0
   3
   6
   9
  12
```

```
[ ] Inner Loop Finished
    0
⇒ 3
   6
   9
  12
   Inner Loop Finished
   0
   4
   8
  12
  16
   Inner Loop Finished
   Outer Loop Finished
```

```
[ ] #Example 8
    x=0
    while x<5:
        x=x+1
        print(x)
    print("Finished")
```

```
⇒ 1
   2
   3
   4
   5
   Finished
```


Fundamentals of Python- Examples

```
[ ] #Example 9
x=60
if x>=70:
    print("Excellent")
elif x>=60:
    print("Good")
else:
    print("OK")
```

⇒ Good

```
[ ] #Example 10
#Input by Program
print("What is your name?")
x=input()
print('Hello ' + x + "!!")
```

⇒ What is your name?
Siri
Hello Siri!

```
[ ] #Example 11
#Functions

def add(x,y):
    z=x+y
    return z
```

```
[ ] #Example 11
#Functions

def add(x,y):
    z=x+y
    return z

t=add(30,20)
print (t)
```

⇒ 50

```
[ ] #Example 12
x = 5
y = "Juan dela Cruz"
print(x)
print(y)
```

⇒ 5
Juan dela Cruz

```
[ ] #Example 13
x = 25      # the data type of x is integer
print(x)
x = "Juan dela Cruz" # the data type of x is now string
print(x)
```

⇒ 25
Juan dela Cruz

Fundamentals of Python- Examples

```
[ ] #Example 14
x = str(3)    # x will be '3'
print (x + '1')

y = int(3)    # y will be 3
print (y+1)

z = float(3)  # z will be 3.0
print (z+0)
```

```
⇒ 31
   4
   3.0
```

```
[ ] #Example 15 (Get the type)
x = 8
y = "Juan Dela Cruz"
print(type(x))
print(type(y))
```

```
⇒ <class 'int'>
   <class 'str'>
```

```
[ ] #Example 16
x = "Juan dela Cruz"
# is the same as
x = 'Juan dela Cruz'
```

```
[ ] #Example 17
var1 = 2
Var1 = "Pedro"
#Var1 will not overwrite var1

print(var1)
print(Var1)
```

```
⇒ 2
   Pedro
```

```
[ ] #Example 18
#CamelCase
myVariableName = "Pedro"

#Pascal Case
MyVariableName = "Pedro"

#Snake Case
my_variable_name = "Pedro"
```

Fundamentals of Python- Examples

```
[ ] #Example 19
x, y, z = "Orange", "Banana", "Cherry"
print(x)
print(y)
print(z)
```

⇒ Orange
Banana
Cherry

```
[ ] #Example 20
x = y = z = "Orange"
print(x)
print(y)
print(z)
```

⇒ Orange
Orange
Orange

```
[ ] #Example 21
fruits = ["apple", "banana", "cherry"]
x, y, z = fruits
print(x)
print(y)
print(z)
```

⇒ apple
banana
cherry

```
[ ] #Example 22
x = "Python is awesome"
print(x)

a = "Python"
b = "is"
c = "awesome"
print(a, b, c)
```

```
a = "Python "
b = "is "
c = "awesome"
print(a + b + c)
```

⇒ Python is awesome
Python is awesome
Python is awesome

Fundamentals of Python- Examples

```
[ ] #Example 23
x = 20
y = 5
print(x + y)
```

⇒ 25

```
[ ] #Example 24
x = 5
y = "John"
print(x, y)
```

⇒ 5 John

```
[ ] #Example 25
'''
This is a multiple line
comments
'''
print('Hello World')
```

⇒ Hello World

```
[ ] #Example 26
a=5
b=7
c=a+b
print ("c = ",c)
```

```
d='x'
e='y'
print (d+e)
```

```
f="Hello"
g="World"
```

```
l=f + " " + g
print (l)
```

⇒ c = 12
xy
Hello World

```
[ ] #Example 27
#Loops
for i in range (5):
    print (i)
    print("Finished")
```

⇒ 0
Finished
1
Finished
2
Finished
3
Finished
4
Finished

```
[ ] #INDENTATION MATTERS!
#Example 28
#Loops
for i in range (5):
    print (i)
    print("Finished")
```

⇒ 0
1
2
3
4
Finished

Fundamentals of Python- Examples

```
[ ] #Accepting Input 1
    print("Enter your name: ")
    x=input()
    print ("Hello "+ x + " !")
```

```
⇒ Enter your name:
   John
   Hello John !
```

```
#Accepting Input 2
print("Enter a number: ")
x=input()
y=int(x)
y=y**2
print("The sum of the square of " + str(x) + " is "+ str(y))
```

```
Enter a number:
2
The sum of the square of 2 is 4
```


Activity 1

- Group yourselves into 3.
- Convert your existing Java code into an equivalent Python code.

```
1 public class ClassProgramTracingQuiz2 {  
    public static void main (String [] args){  
        double numberOfHoursWorked=40;  
        double ratePerDay=500;  
        double tax=0.20, taxDeduction;  
        double grossSalary, netSalary;  
  
        grossSalary=numberOfHoursWorked*ratePerDay;  
        taxDeduction=grossSalary*tax;  
        netSalary=grossSalary-taxDeduction;  
  
        System.out.println("The gross salary is "+grossSalary);  
        System.out.println("The net salary is "+netSalary);  
    }  
}
```

Activity 2

- Group yourselves into 3.
- Convert your existing Java code into an equivalent Python code.

```
1 import java.util.Scanner;

    public class IfConditionalStatement1 {
        public static void main (String [] args)
        {
            int grade;
            Scanner input=new Scanner (System.in);
            System.out.println("Enter a grade: ");
            grade=input.nextInt();
            if (grade>75)
            {
                System.out.println("You passed");
            }
            else
            {
                System.out.println("Sorry, you failed.");
            }
        }
    }
```